The Price of Compliance: Emission-Mask-Constrained Adversarial Attacks on Deep Learning Spectrum Sensing in the 5.9 GHz ITS Band

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Abstract

Deep learning (DL) classifiers are increasingly proposed for spectrum sensing in Vehicle-to-Everything (V2X) networks, and are known to be vulnerable to adversarial perturbations. Prior wireless adversarial-ML evaluations, however, either perturb receiver-side features directly—a threat no radio transmitter can realize—or transmit without regard for spectrum rules. This paper formalizes and quantifies the attacker that actually exists in the 5.9 GHz ITS band: a *compliant attacker* that respects its regulatory allocation and emission mask while optimizing a waveform-domain perturbation end-to-end through the attacker-to-receiver channel and the receiver's STFT front-end. We instantiate the threat model on a post-FCC coexistence sensing task in a 20 MHz window straddling the 5895 MHz boundary: C-V2X PC5 and legacy IEEE 802.11p are co-channel in the ITS band (the transition-period problem), with U-NII-4 Wi-Fi below the boundary. On TR 37.885-inspired urban, highway, and rural channels, an unconstrained ("genie") attacker reaches 20% conditional attack success at −44 dB relative received power, while the mask-compliant attacker needs −37 to −39 dB: an *emission-mask price of compliance of 5–7 dB* (untargeted) and 16–18 dB for targeted "cloaking" of an active transmitter as noise. The absolute

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regime is the more alarming finding: an *undefended* sensing CNN is broken by a rule-abiding transmitter operating tens of dB below the victim signal level (96.7% conditional ASR at -10 dB). We also translate the legacy feature-space threat model into physical power, showing that receiver-internal ϵ-balls describe a perturbation that mostly cannot survive its own physical realization, while a purpose-optimized waveform at the same implied power is far stronger. As the matched defense, mask-matched adversarial training shifts the compliant-attacker threshold by $+14$ dB under the evaluation attack (PGD-10)—but a converged adaptive attack (PGD-50 with random restarts) recovers most of that margin, leaving $+6.7$ dB of honestly adaptive robustness on the canonical defense seed (replicated across three defense training seeds: $+9.8 \pm 2.8$ dB for AT, $+11.5 \pm 1.8$ dB for TRADES—the 3.5–5.5 dB seed spread is itself a certification finding), and the decomposition attributes the gap to evaluation step-starvation rather than restart luck. These conclusions are stress-tested beyond the prototype protocol: three training seeds move the compliant threshold by at most 1.1 dB; replacing the WiFi class with real over-the-air 802.11 captures (recorded at 5.24 GHz in U-NII-1, the OFDM family adjacent to U-NII-4) replicates the compliant MEAP within 1.8 dB while exposing a decision-margin fragility that synthetic-only evaluation hides entirely; and a second victim architecture (early-fusion ResNet) shifts the threshold by 11.6–15.1 dB, with surrogate-model transfer costing the attacker 11–23 dB. A power-amplifier reality check then shows what compliance costs beyond floating point: the optimized attack waveform has the highest PAPR on the air (11.1 dB vs. 6.3–9.0 dB for the classes it impersonates), and a Rapp-model PA only keeps its post-amplifier spectrum inside a -40 dBr mask at 13–15 dB backoff—while the attack itself remains effective at any backoff, making post-PA regrowth statistics an RF-side tell against the naive attacker; a PA-aware re-optimization closes even that tell while gaining 1.2 dB. The mask idealization itself is then validated against the real ETSI EN 302 571 Table-7 unwanted-emissions template: the compliant MEAP moves by only $+0.4$ dB, so the price-of-compliance headline is not an artifact of the self-defined mask shape. Finally, a disclosed-parameter link-budget layer translates

the sensing attack into fleet-scale harm: at the targeted 20% false-IDLE operating point a rule-abiding attacker within the 33 dBm ITS EIRP cap subtracts 9.5 pp of BSM packet-reception ratio at 100 m legit-link distance (∼95 extra lost BSMs per 1000 transmitted). We propose Minimum Effective Attack Power (MEAP) and Price-of-Compliance (PoC) as physically grounded, transferable security metrics, and release the complete framework (waveform generators, channels, attack, defenses, checkpoints, result JSONs with embedded seeds and configurations) for reproduction.

Keywords: V2X, spectrum sensing, adversarial machine learning, 5.9 GHz, C-V2X, IEEE 802.11p, emission mask, physical-layer security

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1. Introduction

The 5.9 GHz ITS band is in the middle of the most consequential spectrum decision of its history. In FCC 20-164 (2020), the Commission reallocated the band: the upper 30 MHz (5895–5925 MHz) is dedicated to C-V2X, while the lower 45 MHz (5850–5895 MHz) was opened to unlicensed U-NII-4 operation [12]; the 2024 Second Report and Order completed the transition rules and set the DSRC sunset in motion [13]. The transition is no longer prospective: the final rules took effect in February 2025 with a two-year DSRC sunset, more than fifty waivers already authorize C-V2X operation in the band [14], and the industry roadmap projects volume deployment of direct-mode V2X radios in production vehicles [15]. During and after this transition, coexistence sensing—determining *which* technology is transmitting, not merely whether energy is present—becomes a safety-relevant function: roadside units and vehicles must distinguish C-V2X PC5, legacy 802.11p still on the road, and unlicensed Wi-Fi spilling up to the boundary. Deep learning classifiers are the natural candidate for this four-way technology recognition, following their success in wireless signal classification [1, 2].

The same classifiers, however, inherit the adversarial fragility of deep networks. A large literature demonstrates that wireless DL classifiers can be fooled by small, deliberate perturbations [3, 6]. Yet the threat models in that literature have a physical gap. Feature-space attacks edit the receiver's input (IQ frames or spectrograms) directly—an adversary with that access does not need adversarial examples at all. Over-the-air and channel-aware attacks [4, 5, 8] closed part of the gap by transmitting real waveforms through modeled channels, and recent work has targeted wideband spectrum sensing directly [7]. But none of these asks the question that every real attacker in the 5.9 GHz band must answer: *the attacker must obey the very spectrum rules it is attacking.* A C-V2X device may only transmit in its 10 MHz ITS allocation under an emission mask (ETSI EN 302 571 [18] in Europe; FCC rules in the US); a U-NII-4 device must stay below 5895 MHz. An attack that requires illegal emissions is not an attack on the system—it is a regulatory violation that conventional spectrum enforcement already handles.

This paper therefore poses and quantifies the *compliant-attacker* threat model: a spectrum-rule-abiding adversary that transmits only within its allocation, under an emission mask, with bounded power, while optimizing its waveform end-to-end through the physical channel and the receiver's front-end. The question that determines whether mask compliance is itself a defense is quantitative: *how much attack power does compliance cost?* We answer it with two new metrics—Minimum Effective Attack Power (MEAP) and Price of Compliance (PoC)—and with a fully differentiable waveform→channel→STFT→CNN attack chain whose perturbations are projected, at every optimization step, onto the physically feasible set.

Our contributions are:

1. **Threat model.** The compliant attacker: an emission-mask and allocation constraint on waveform-domain adversarial perturbations, to our knowledge the first such constraint in the wireless adversarial-ML literature (Table 1).
2. **Task.** A post-FCC coexistence sensing benchmark in a 20 MHz window straddling the 5895 MHz boundary: block-exact 802.11p and U-NII-4 gen-

erators and a rate-matched C-V2X PC5 generator, co-channel 802.11p/PC5 in the ITS allocation (the transition problem), with an honest energy-feature baseline showing where deep learning is and is not needed.

3. **Attack.** A differentiable projected-gradient attack over the waveform, with out-of-band nulling, an in-band PSD cap, and an explicit power budget defined as the physical attack-to-signal power ratio (PSR).

4. **Metrics.** MEAP and PoC: power-domain, attacker-agnostic robustness measures suitable for certification-style reporting.

5. **Findings.** On TR 37.885-inspired channels: compliance costs the attacker $\sim$5–7 dB (untargeted) and 16–18 dB (targeted cloaking); an undefended CNN is nevertheless broken by a compliant attacker at deeply negative PSR; the legacy feature-space ϵ-ball assumes a hidden power budget we make measurable; and mask-matched adversarial training shifts the compliant-attacker threshold by +14 dB under the evaluation attack (+6.7 dB against the converged adaptive attack on the canonical defense seed; $+9.8 \pm 2.8$ dB across three defense seeds; Section 5.6) at no clean-accuracy cost. Compliance *delays* the attack; so does the defense; neither prevents it.

Scope and honesty. All attack numbers in this paper are prototype-scale (three training seeds for the undefended headline, two attack seeds, 300 evaluation windows, PGD-10) and are labeled as such; the full protocol (PGD-50, extended grid) is specified in the released repository; rural, PGD-50, a second attack seed, the mask-cap sensitivity, the defense study, the feature-space comparison, real over-the-air captures, a second victim architecture, and surrogate-model transfer are included here. Every number traces to a result JSON with embedded configuration. The undefended model is the subject of Sections 5.3–5.4; the defense study (Section 5.5) quantifies how much of the threat adversarial training removes.

2. Related Work and Threat-Model Positioning

DL spectrum sensing and ITS-band recognition. Deep learning radio signal classification was established at scale by O'Shea et al. [1]. For the ITS band specifically, Girmay et al. [2] built CNN-based technology recognition and traffic characterization for coexisting wireless technologies—the coexistence problem our task generator mirrors. These works establish capability, not robustness.

Adversarial attacks on wireless DL. Sadeghi and Larsson [3] demonstrated white-box attacks on modulation classifiers in feature space. Liu et al. [6] attacked spectrum sensing in cognitive-radio IoT, again at the feature level. Over-the-air reality was first confronted by Kim et al. [4] and matured into channel-aware attacks [5], which optimize the transmit waveform through a modeled channel; RadioShock [8] demonstrated OTA attacks with channel-state estimation, and Zheng et al. [7] brought targeted attacks to wideband spectrum sensing. Defense-side, Zhao et al. [9] proposed statistical detection of adversarial spectrum attacks, and Croce and Hein [10] established reliable attack ensembles for robustness evaluation. In the standards world, Habler et al. [11] mapped adversarial-ML threats onto O-RAN architectures—the deployment context our metrics target.

Primary-user emulation: the pre-DL ancestor. Constraining the *attacker* rather than its perturbation has a long history in a neighboring community. Primary-user-emulation attacks (PUEA)—a malicious node transmitting so that energy-detection sensors believe the primary user is present (denial of service) or absent (spectrum theft)—were formalized by Anand et al. [16] and remain an active detection literature [17]. PUEA and the compliant attacker share one premise—the adversary transmits a real waveform—but differ in everything downstream: the victim (an occupancy threshold vs. a deep classifier), the mechanism (statistical mimicry of the primary's signal and power geometry vs. a gradient-optimized perturbation through the receiver's front-end), the constraint set (attacker power and distance, but no allocation or emission-mask constraint, vs.

Table 1: Threat-model comparison. "Mask" = regulatory allocation + emission-mask constraint on the transmitted adversarial perturbation. PUEA = primary-user emulation, the pre-DL ancestor (Section 2). "Power sweep": *fixed* = single power level; *PNR* = accuracy-vs-PNR sweeps with no threshold-in-power metric; *MEAP* = threshold-in-power robustness metric (this work).

Work	Waveform	Channel	Mask	Power sweep	Task
Sadeghi & Larsson [3]	–	–	–	–	modulation
Liu et al. [6]	–	–	–	–	spectrum sensing
Anand et al. [16] (PUEA)	✓	–	–	fixed	occupancy
Kim et al. [4]	✓	✓	–	fixed	modulation
Kim et al. [5]	✓	✓	–	PNR	modulation
Zheng et al. [7]	✓	✓	–	fixed	wideband sensing
RadioShock [8]	✓	✓	–	fixed	communication
This work	✓	✓	✓	**MEAP**	**ITS coexistence**

the regulatory feasible set here), and the reported quantities (occupancy error rates vs. power-domain robustness metrics). The compliant-attacker threat model is not a rediscovery of PUEA; it is its adversarial-ML descendant, inheriting the "real waveform, real power budget" premise while replacing the victim, the optimization, and the measurement.

Positioning. Table 1 positions the compliant attacker. The differentiator is not "channel awareness" (present in [5, 8]) nor the sensing task (present in [6, 7]); it is the *regulatory constraint set*: allocation, emission mask, and an explicit power budget swept as the independent variable. To our knowledge no prior work constrains a wireless adversary to its spectrum rules, and none reports a power-domain price-of-compliance analysis. This matters because compliance is the one attacker property that is *verifiable by regulators*: an attacker that must obey the mask is within the reach of conformance testing, which makes the resulting robustness metrics deployable (Section 6).

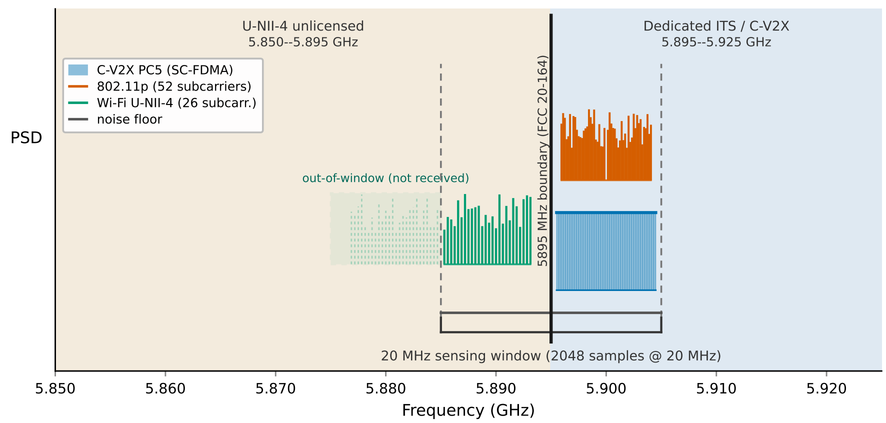

Figure 1: Post-FCC 5.9 GHz band plan as seen by the boundary-straddling sensing window (stylized PSD, not to scale). Shading marks the two regulatory blocks created by FCC 20-164: unlicensed U-NII-4 below the 5895 MHz boundary and the dedicated ITS/C-V2X allocation above it. C-V2X PC5 (flat-topped SC-FDMA block) and IEEE 802.11p (52 raised subcarrier stems) are deliberately co-channel in the lowest 10 MHz ITS channel; the Wi-Fi class is the partial 26-subcarrier view of a U-NII-4 channel centered 10 MHz below the window, whose out-of-window half (dashed ghost) is removed by the receiver's anti-alias filter.

3. System Model

3.1. Post-FCC coexistence sensing task

The sensing receiver observes a 20 MHz complex-baseband window (2048 samples at $F_S = 20\,\mathrm{MHz}$, i.e., 102.4 µs) centered on the 5895 MHz boundary, covering 5885–5905 MHz (Fig. 1). This boundary-straddling placement is the physically interesting vantage point: the upper half contains the ITS allocation (5895–5905 MHz is the lowest C-V2X channel), the lower half the upper edge of U-NII-4 unlicensed operation. The classification task is four-way (Table 2):

Two design decisions deserve justification. First, *802.11p and PC5 are co-channel* in the ITS allocation: during the multi-year DSRC sunset, roadside infrastructure must classify both technologies sharing the same spectrum, and region-energy features reliably cannot separate them—our logistic-regression baseline on region-energy and envelope features reaches only 77.75% on this pair, while the CNN reaches 100% (four-class baseline: 88.62% vs 100%). The

Table 2: Post-FCC band plan as seen by the sensing window (frequencies relative to the 5895 MHz boundary). PC5 and 802.11p are deliberately co-channel: the transition-period coexistence problem.

Class	Standard basis	Occupied (MHz)	Generation
C-V2X PC5	Rel-14 SC-FDMA†	$[+0.5, +9.5]$	contiguous DFT-s-OFDM
802.11p	802.11-2012 OFDM	$[+0.94, +9.06]$	52-sub IFFT+CP, DC nulled
Wi-Fi U-NII-4	802.11 20 MHz OFDM	$[-9.69, -1.88]$	26 in-window subcarr.
Noise	–	full band	complex AWGN

†rate-matched stylization: 450×20 kHz (9 MHz = Rel-14 occupied BW); SCS/DMRS not modeled (Sec. 7).

DL-necessity claim is thus confined to what physics cannot do, unlike prior work where classes are region-separable and "deep learning needed" claims do not survive an energy baseline. Second, *Wi-Fi is region-separable from the ITS pair* (it lives below the boundary); this is correct post-FCC physics, and we state it rather than engineer it away.

The waveform generators are block-exact where the standards allow it at $F_S = 20$ MHz: 802.11p [21] uses 52 subcarriers at 156.25 kHz spacing with a 1.6 µs cyclic prefix and a nulled DC subcarrier; the U-NII-4 class is the physically correct partial view of a 20 MHz 802.11 channel (312.5 kHz spacing, 0.8 µs guard interval) whose center lies 10 MHz below the window—only the 26 in-window subcarriers are generated, as the receiver's anti-alias filter removes the rest. PC5 is a rate-matched stylization of Rel-14 [20]: 450 subcarriers at 20 kHz (the same 9 MHz occupied bandwidth as 50 PRBs at 15 kHz), chosen so the block is an exact integer at $F_S = 20$ MHz; single-carrier structure and PAPR are preserved (measured: 6.26 dB PC5 vs 8.77/8.69 dB for the OFDM classes, matching the committed benign-PAPR statistics in our PA reality-check data). All data symbols are QPSK; pilots and preambles are not modeled.

3.2. Channel model

Channels are simplified, TR 37.885-inspired [19] block-fading tapped-delay models: highway (120 km/h, Rician $K = 10$ dB first tap, 5 taps, ∼0.4 µs spread), urban (50 km/h, Rayleigh, 6 taps, ∼1.3 µs spread, UMa-flavored), and rural (30 km/h, $K = 5$ dB, 3 taps). Block fading is justified even at highway speed: the maximum Doppler (∼656 Hz at 5.9 GHz) gives a decorrelation time of ∼0.55–0.76 ms, far exceeding the 102.4 µs window. The victim link and the attacker-to-receiver link draw independent channel realizations. Deviations from the spec's CDL profiles (first-tap-only Rician, UMa-flavored urban spread) are disclosed in Section 7.

3.3. Receiver front-end and classifier

The receiver computes a complex STFT (256-point, 128 hop, Hann window, two-sided and frequency-shifted) yielding 256×15 frames, from which two streams are derived: log-magnitude (z-scored, clamped) and the per-bin temporal phase difference (a phase-vocoder-style frequency-offset estimate). Both streams feed a dual-stream Inception-Time CNN (86,052 parameters), carried over from our prior work [22] for architectural comparability; a magnitude-only variant (40,980 parameters) serves as the pre-registered stream ablation. The entire chain is implemented in PyTorch and is differentiable end-to-end from time-domain waveform to logits—the enabler of the waveform-domain attack. Training uses the prior work's recipe (label smoothing 0.1, TF-CutMix, Gaussian augmentation, Adam 5e-4, cosine schedule) on 3,200 training windows (800 held-out validation windows) with mixed scenarios and SNR drawn uniformly from [5, 25] dB.

4. The Compliant Attacker

4.1. Threat model

The adversary is a C-V2X device transmitting in its own ITS allocation $[0, +10]$ MHz relative to the boundary. Its perturbation $\delta \in \mathbb{C}^N$ must satisfy, *at its transmit port*: (i) *allocation*: $\mathrm{supp}(\hat{\delta}) \subseteq \mathcal{B}_a$ (out-of-band FFT bins nulled); (ii) *emission mask*: $|\hat{\delta}_k|^2 \leq M_k$ in-band (a flat PSD cap at 2× uniform budget sharing—an idealization stricter than real in-band regulations, whose sensitivity

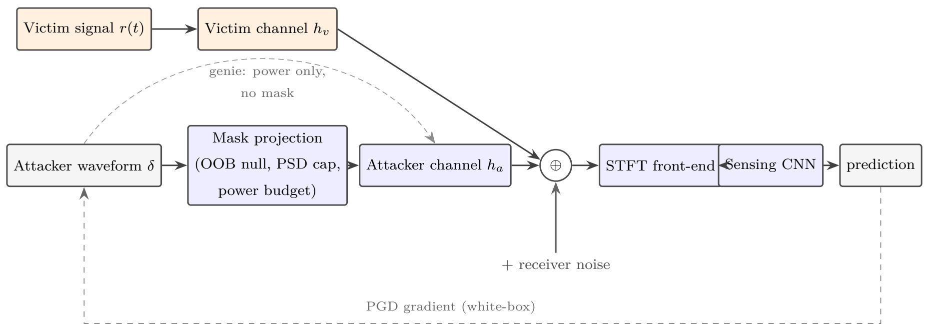

Figure 2: Threat model of the compliant attacker. The attacker optimizes a waveform-domain perturbation δ end-to-end through its own channel h_a, the receiver's STFT front-end, and the sensing CNN; the dashed return path carries the white-box PGD gradient. The *compliant* variant projects δ onto the feasible set (out-of-band null, in-band PSD cap, transmit-power budget) at every optimization step, while the *genie* variant bypasses the mask projection and obeys only the power budget. The victim signal $r(t)$ arrives through an independent channel h_v; the junction also adds receiver noise.

we report); and (iii) *power*: $\|\delta\|_2^2 \leq P_a$, with the physical budget expressed as PSR (dB) $= 10\log_{10}(E_{\text{attack}}/E_{\text{clean rx}})$ over equal-length windows—the ratio of adversarial transmit power to received victim-signal power. A *genie* baseline obeys only (iii): the classical unrestricted upper bound. The attacker knows the victim model and its own attacker-to-receiver channel (white-box with CSI); *worst-case disclosure*: the optimization also uses the exact received realization (signal + noise), so all reported ASRs are upper bounds—the direction favorable to the attack. Section 5.3 quantifies how this degrades under CSI error and with no CSI at all. Figure 2 summarizes the attack chain and the two attacker variants.

4.2. Feasible set and projected-gradient attack

$$\mathcal{F} = \{\delta \in \mathbb{C}^N : \text{supp}(\hat{\delta}) \subseteq \mathcal{B}_a,\ |\hat{\delta}_k|^2 \leq M_k,\ \|\delta\|_2^2 \leq P_a\} \tag{1}$$

The attack optimizes δ by PGD through the differentiable chain $\delta \to h_a * \delta \to +r(t) \to \text{STFT} \to \text{CNN} \to \mathcal{L}$, with loss $\mathcal{L} = -\text{CE}$ (untargeted) or $\text{CE}(\cdot, y_{\text{target}} = \text{Noise})$ (targeted "cloaking": make an active transmitter look like an idle band). After each step, δ is projected onto $\mathcal{F}$: FFT-domain out-of-band nulling, per-bin PSD capping (Parseval-corrected: $\sum_k |\hat{\delta}_k|^2 = N\|\delta\|_2^2$), and total-power rescaling. The projection is exactly idempotent and enforces the budget to machine precision (verified: post-projection power ratio 1.000000, out-of-band fraction 10^{-14}). Compliance is defined at the transmit port; the finite sensing window spreads the *received* adversarial component to $\sim -33.5\,\text{dB}$ out-of-band fraction—a receiver artifact, not a transmit violation.

4.3. Physical security metrics

Raw ASR conflates attack success with the classifier's base error; we report *conditional ASR* over clean-correctly-classified windows, and *robust accuracy* over all windows. Two power-domain metrics summarize the threat: *MEAP* (Minimum Effective Attack Power) is the PSR at which conditional ASR crosses 20% (linear interpolation, with explicit censoring flags when the grid does not resolve the crossing); *Price of Compliance* is $\text{PoC} = \text{MEAP}_{\text{compliant}} - \text{MEAP}_{\text{genie}}$: the power a rule-abiding attacker must additionally spend to achieve the same attack success. PoC is budget-matched, attacker-agnostic, and—unlike ASR at a fixed ϵ—directly interpretable in link-budget terms.

5. Experimental Evaluation

5.1. Protocol and scale

Unless stated otherwise, every experiment below uses the same evaluation contract: a frozen victim model, a fixed set of active-transmission windows (PC5, 802.11p, WiFi), per-sample attacker channels drawn from the scenario model, and the PSR grid of Table 3; conditional ASR is measured only over windows the victim classified correctly before the attack, and every MEAP carries an explicit censoring flag when the grid does not resolve it. Every conditional-ASR cell is stored with its success count and eligibility, and headline MEAP/PoC numbers carry Wilson binomial 95% confidence intervals (sampling noise at fixed model/protocol; model-seed spread is reported separately). The honesty box records the prototype scale; the three-seed statistics, PGD-50 convergence, adaptive (multi-restart) evaluation of the defenses, and sensitivity checks in Section 5.3 quantify how far the prototype numbers generalize.

Table 3: Protocol honesty box. Prototype-scale numbers in this paper use the first row; the full protocol is released with the repository.

Setting	Value (prototype)
Training	seeds 42/123/456 (undefended headline), 3,200 train / 800 val windows, SNR [5,25] dB; defenses s
Attack	seed 7, 300 active-tx windows, PGD-10, $\alpha = 0.25$
Adaptive attack	PGD-50, 5 restarts (zero-init + 4 random), best-of-R per sample; 3 defense seeds
PSD cap margin	2.0 (flat-cap idealization; relaxing to 10 moves PoC by -0.9 dB)
PSR grid	-45 to $+10$ dB, 12 points
MEAP threshold	20% conditional ASR
Confidence	Wilson 95% binomial CIs on every ASR cell (results/w12_confidence_intervals.json)

5.2. Clean sensing performance

The dual-stream CNN and the magnitude-only ablation both reach 100% validation accuracy (800 held-out windows); the energy-feature logistic regression baseline reaches 88.62% (four-class) and 77.75% on the co-channel PC5-vs-802.11p pair, where the CNN is error-free. Across the three training seeds (42/123/456) the picture is stable: clean accuracy is 100% for all seeds, the LR baseline moves within one point (88.6/88.1/87.8%), and the LR pair accuracy spreads 74.0–77.8% (mean 76.3%) while the CNN pair stays error-free throughout. The phase difference stream contributes nothing (dual = mag-only), retiring the "phase-aware" framing of our prior work [22]—an honest negative result, pre-registered as an ablation. An SNR sweep below the training range bounds the noise-floor regime honestly: the CNN's advantage over the energy baseline persists down to 0 dB SNR (71.8% vs 61.3% at $[0, 10]$ dB, 63.2% on the hard pair) and vanishes at negative SNR (24.3% vs 25.0% at $[-10, 0]$ dB)—both models starve; deep learning does not rescue a task with no signal. (The first version of this sweep drew its evaluation baseband from the

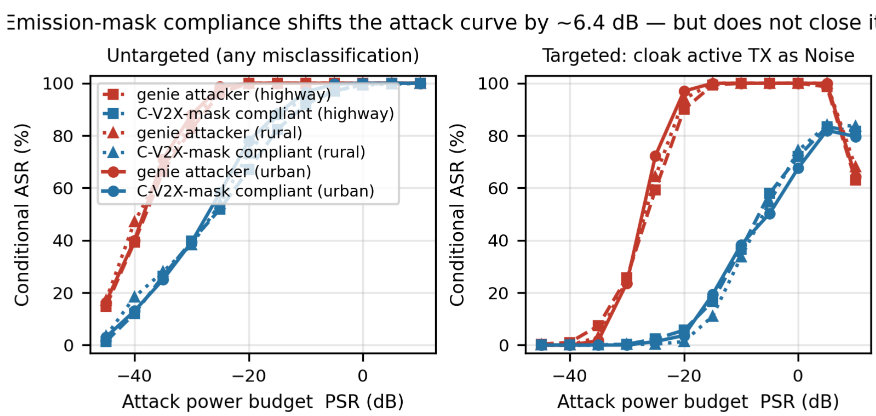

Figure 3: Conditional ASR versus attack power budget (PSR, dB) for genie vs emission-mask-compliant attackers, untargeted and targeted-to-Noise modes, urban, highway, and rural channels (prototype protocol of Table 3). Compliance shifts the attack curve right by ∼5–7 dB (untargeted) but does not close it; both attackers break the undefended CNN at deeply negative PSR.

training seed's RNG stream; a hash-level audit found 2/1000 realized waveforms bit-identical to training-corpus members. The numbers above use a fresh evaluation seed verified at 0/1000 overlap; every accuracy moved by at most 1.3 pp, within sampling noise at n=1000—the leak was real but immaterial, and both the audit and the clean re-run are released.)

5.3. The price of compliance

Four findings emerge (Fig. 3, Table 4). **(F1) Undefended CNNs are catastrophically brittle in absolute terms.** The genie attacker reaches 20% conditional ASR at −44 dB—four orders of magnitude below the received signal—and the *compliant* attacker at −37 dB. At −10 dB (an attacker ten times weaker than the victim signal), the compliant attacker achieves 96.7% (urban) / 92.0% (highway) conditional ASR, and 39.3% (urban) / 39.7% (highway) even at −30 dB. **(F2) Compliance costs ∼5–7 dB** for generic misclassification, consistently across urban, highway, and rural channels—the price is a property of the mask, not the channel. **(F3) Cloaking is the costlier attack:** forcing the specific "Noise" label costs the compliant attacker 16–18 dB more than the genie

Table 4: Physical summary metrics (conditional-ASR threshold 20%; prototype protocol of Table 3; all MEAPs resolved within the grid, no censoring). Training-seed stability (seeds 42/123/456, urban untargeted): compliant MEAP $-37.1/-38.1/-37.9$ dB (mean -37.7, spread 1.1 dB); genie MEAP at or below the grid floor for 2 of 3 seeds, so the per-seed PoC (7.1/6.9/7.1 dB) is a lower bound for those seeds.

Scenario	Mode	MEAP genie	MEAP compliant	Price of Compliance
urban	untargeted	−44.2 dB	−37.1 dB	7.1 dB
highway	untargeted	−43.9 dB	−37.2 dB	6.7 dB
rural	untargeted	−44.5 dB	−39.1 dB	5.4 dB
urban	targeted-noise	−30.8 dB	−14.8 dB	16.0 dB
highway	targeted-noise	−31.5 dB	−14.2 dB	17.4 dB
rural	targeted-noise	−31.1 dB	−13.0 dB	18.1 dB

(0.4% or less conditional ASR below -25 dB; 38.3%/36.7% at -10 dB; 68%/72% at parity). Target specificity, not generic misclassification, is where the mask bites. **(F4) Overshoot at high power:** at $+10$ dB the targeted attack's Noise-specific success *decreases* (genie 63–68%, compliant 80–84%) while untargeted damage saturates—the perturbation overshoots into other wrong classes (robust accuracy is 8–14%), a real phenomenon for certified-decision pipelines that key on specific labels.

Robustness of the metrics. Three checks bound the main threats to validity of Table 4. *Attack-strength convergence:* PGD-50 (versus PGD-10) shifts the compliant MEAP by only -1.7 dB (to -38.8 dB; its genie MEAP is censored at that grid's floor) and raises mid-grid conditional ASR substantially (up to $+37$ pp at -25 dB)—the zero-initialized, restart-free PGD-10 protocol therefore reports materially *conservative* attack numbers, and the PGD-10 MEAPs are lower bounds. *Seed stability:* a second attack seed (11) reproduces the urban untargeted PoC at 6.9 dB versus 7.1 dB (0.3 dB spread; MEAPs move by ≤ 1 dB). *Mask-cap idealization:* relaxing the flat in-band cap from $2\times$ to $10\times$ uniform sharing moves the compliant MEAP by only -0.9 dB and the PoC from 7.1

to 6.2 dB—the Price of Compliance is driven by the *allocation* constraint (the out-of-band null), not by how strictly in-band flatness is enforced. *Attacker knowledge:* re-running the compliant attack with imperfect CSI (additive error of 0.3 relative magnitude) costs at most 2.7 pp of conditional ASR across the measured grid—the attack is robust to CSI error; but with *no* CSI (optimizing against an independent draw from the same scenario, evaluated on the true channel) the compliant attack's 20% ASR threshold moves from -37 dB to ≈ -16 dB, a $\sim$21 dB penalty: channel-realization knowledge, not channel statistics, is what makes the white-box threat catastrophic. *Training-seed statistics:* retraining the victim from scratch on seeds 123 and 456 (identical recipe) and re-running the full urban untargeted grid gives compliant MEAPs of -38.1 and -37.9 dB against seed 42's -37.1 dB—a 1.1 dB total spread, with clean accuracy 100% and the CNN pair error-free on every seed. The genie MEAP falls at or below the -45 dB grid floor for the two new seeds, making their PoCs lower bounds (6.9 and 7.1 dB); the ordering genie < compliant threshold, and the $\sim$7 dB price of compliance, are invariant to the training seed.

5.4. What the legacy feature-space threat model silently assumes

A log-magnitude perturbation of $\pm\epsilon$ (in z-scored units) in the receiver's spectrogram has no direct physical meaning. It can be given one: a log-magnitude change d on bin k corresponds to the complex STFT-domain addition $\hat{\Delta}_k = \hat{Z}_k(10^d - 1)$ (magnitude scaling, phase preserved), and the hann/50%-overlap STFT is invertible (interior reconstruction error 8.4×10^{-7}), so the same edit implies a *received* waveform perturbation whose power we can measure ($\mathrm{PSR}_{\mathrm{eq}}$). We attack the magnitude input with an ℓ_∞ PGD ($\epsilon \in \{0.1, 0.3, 1.0\}$) and translate the perturbation to the waveform domain. At $\epsilon = 1.0$ (one standard deviation of the log-magnitude distribution) the feature-space attack achieves 71.3% conditional ASR *as evaluated in the legacy setting* (the perturbed spectrogram is fed directly to the CNN), while its implied received power is -24 dB (median). Feeding the actual waveform $x + \delta_{\mathrm{rx}}$ through the receiver's true front-end realizes only 14.0%: the hann-window round trip realizes roughly half of the intended per-bin log-change, and the receiver noise absorbs the rest. By contrast, *transmitted* attacks optimized for that same power (-24 dB) achieve 99.3% (genie) and 62.3% (compliant). At $\epsilon = 0.3$ the legacy attack achieves 1.7% intended (0% realized) at an implied -34.4 dB, and at $\epsilon = 0.1$ nothing (0%) at -44.0 dB. Three lessons: (i) the feature-space ϵ-ball hides a power budget that is invisible in ϵ-units; (ii) a feature-space perturbation is not a physically consistent waveform—its own physical realization mostly fails, so legacy feature-space ASRs overstate realizable attacks at the implied power; (iii) an adversary that can actually transmit is far better served by optimizing the waveform directly (as in this paper) than by hoping an ϵ-ball transfers. Comparisons of ASR across threat models are meaningless without the $\mathrm{PSR}_{\mathrm{eq}}$ translation.

5.5. Defenses: mask-matched adversarial training

The natural defense is adversarial training inside the same feasible set: during training, each minibatch is (with probability 0.5) adversarially perturbed by a 5-step mask-compliant waveform PGD attack at a PSR drawn uniformly from $[-25, -5]$ dB, with a fresh attacker channel, generated against the current model and added through that channel; training otherwise follows the standard recipe. The defended model retains *100%* clean validation accuracy. Against the identical attack protocol (urban, untargeted, seed 7, PGD-10): the genie MEAP shifts from -44.2 to -37.5 dB ($+6.8$ dB), the *compliant* MEAP from -37.1 to -23.1 dB ($+14.0$ dB), and the Price of Compliance doubles from 7.1 to 14.4 dB. Point-wise, the compliant attacker's conditional ASR at -25 dB collapses from 58.3% to 16.7%, and at -10 dB from 96.7% to 55.3%; targeted cloaking at -10 dB falls from 38.3% to 15.0%. Compliance and adversarial training thus compound: each buys roughly 7–8 dB independently, and together they push the compliant attacker's 20%-ASR threshold from -37 dB to -23 dB. The defense has an honest boundary: robustness is confined to the trained power range—at PSR parity and above the attack still succeeds (94.7% at 0 dB), and attacks far below the training range partially transfer. Adversarial training *delays* the attack exactly as compliance does; neither closes it, and the certification-style metric (MEAP) is what quantifies both.

TRADES-style ablation. A TRADES variant ($\beta = 6$, the same mask-matched generator, probability 0.5, PSR $[-25, -5]$ dB, seed 42) replaces the adversarial-view CE with the clean/adversarial KL trade-off objective. It buys *more* threshold shift on the compliant axis (mask MEAP −19.1 dB versus standard AT's −23.1 dB; PoC 18.6 versus 14.4 dB) at a 0.75 pp clean-accuracy cost (99.25% versus 100%). Standard AT remains the zero-clean-cost choice; TRADES is the stronger option when a small clean cost is acceptable—the same trade-off shape documented in the vision literature, here in a power-budget metric.

Cross-scenario transfer. The AT model was trained on urban/highway batches only; evaluated on the rural scenario it retains most of its robustness (compliant MEAP −21.9 dB versus −23.1 dB urban; genie −35.3 versus −37.5 dB; PoC 13.4 versus 14.4 dB): the defense transfers to an unseen channel scenario at a ∼1.2 dB cost.

5.6. Adaptive attacks: step-starved evaluations overstate defenses

The +14 dB headline above is measured under PGD-10, the same budget as the undefended baseline. That is not how a certification-style claim should be evaluated: a defense must hold against a *converged* adversary. We therefore re-evaluated the undefended control, the AT model, and the TRADES model under the strongest attack in our arsenal: PGD-50 with five restarts (restart 0 is the deterministic zero-init shared with every number above; restarts 1–4 are projected complex-Gaussian inits at a random fraction $U[0.2, 1]$ of the per-sample budget), keeping the per-sample best attack objective. Table 5 shows the result. Three findings:

1. **The defense margin halves.** Under the adaptive attack the compliant MEAP of the AT model moves from −23.1 to −32.7 dB and TRADES from −19.1 to −29.6 dB, while the undefended control moves only from

-37.1 to -39.4 dB. The honest defense margins are $+6.7$ dB (AT) and $+9.9$ dB (TRADES), not the $+14.0/+18.0$ dB the PGD-10 protocol suggested. Both defenses still buy real margin, and TRADES remains the stronger one—but every defense claim in this paper should be read against the adaptive column.

2. **The gap is step starvation, not restart luck.** On the AT model, PGD-50 *without* restarts already reaches -31.6 dB (8.5 dB of the 10.4 dB total shift); the four random restarts add only 1.1 dB. The original evaluation was simply not converged—the defended model's loss landscape needs more steps, and a 10-step budget understates the adversary. Evaluation protocol, not defense design, was the dominant error source.
3. **The undefended attack was already near-converged.** The control model loses only 2.3 dB to the adaptive protocol, confirming that the canonical undefended numbers (MEAP -37.1 dB) were not an artifact of a weak attack, and that restarts matter exactly where the defense makes the optimization landscape harder.
4. **Defense-seed replication: the single-seed margin was the weak draw.** We repeated the adaptive evaluation on two additional independently trained defenses per family (AT and TRADES retrained from scratch with seeds 43 and 53; identical recipe; clean accuracy 99.3–100%, so the defense trains reliably). The AT margin is $+6.7/+12.3/+10.3$ dB across defense seeds $\{42, 43, 53\}$ and the TRADES margin $+9.9/+13.4/+11.4$ dB: means $+9.8$ dB (AT, range 5.5 dB) and $+11.5$ dB (TRADES, range 3.5 dB). Three things follow. First, the canonical seed 42 happened to be the *weakest* draw for both families, so the single-seed numbers understate mean robustness. Second, the 3.5–5.5 dB defense-seed spread is 3–5$\times$ the 1.1 dB training-seed spread of the undefended headline—single-seed defense evaluation is not certification-grade, and we label ours accordingly. Third, the ordering TRADES $>$ AT holds on every seed, and both adaptive margins remain far below their PGD-10 values ($+14.0/+18.0$ dB): the step-starvation finding is robust in direction, but its magnitude is seed-

Table 5: Adaptive-attack evaluation (urban, untargeted, attack seed 7, 300 active windows; PGD-50, 5 restarts, best-of-R per sample). The PGD-10 column is the canonical protocol from Table 3; the margin columns are over the undefended control under the SAME attack protocol (undefended adaptive MEAP −39.4 dB). The final column replicates the adaptive attack on three independently trained defenses per family (seeds 42/43/53; mean ± sample std). Wilson 95% CIs on the MEAPs are ±1.3–1.6 dB.

	mask MEAP (dB)			defense margin (dB)		
Victim	PGD-10	PGD-50 R=1	PGD-50 R=5	PGD-10	adaptive (s42)	adaptive (3-seed)
Undefended	−37.1	—	−39.4	—	—	—
AT	−23.1	−31.6	−32.7	+14.0	+6.7	+9.8 ± 2.8
TRADES	−19.1	—	−29.6	+18.0	+9.9	+11.5 ± 1.8

dependent.

The zero-init restart share is itself diagnostic: against the defended models only 16–30% of samples are won by the deterministic restart (restart 0) that the original protocol used exclusively (AT 19–30%, TRADES 16–29%; undefended control 20–40%)—the deterministic protocol was leaving attack strength on the table everywhere, and most where the defense had made the landscape harder. Repro artifacts: results/adaptive_{at,trades,dual}_s7_r5.json (R=5) and adaptive_at_s7_r1.json (R=1 decomposition), each cell with conditional ASR, eligibility, and the win histogram; the defense-seed replication adds results/adaptive_{at,trades}_{d43,d53}_s7_r5.json plus the re-derivation and independent audit in results/w13_defense_seed_replication.json (checked by scripts/w13_check.py, F1–F8).

5.7. *Real over-the-air signals in the loop*

Every result so far—like nearly every result in the wireless adversarial-ML literature—uses synthetic victim signals. We replace the WiFi class in the evaluation set with real over-the-air 802.11 captures (Fontaine et al. dataset, USRP captures at 5240 MHz at three Gent, Belgium locations (a fourth capture site was rejected by the burst gate); DC-removed, resampled to 20 Msps, placed in the unlicensed half of the window, burst-gated by spectral flatness and power; 96 verified active windows; CC BY-NC-SA, redistributed with attribution). Real captures carry their own real propagation channel, so no synthetic victim channel is applied to them; receiver noise is added at the drawn SNR as for every class. Evaluation uses captures from one location (UZ, 74 windows), fine-tuning uses disjoint locations (Rabot/Reep, 22 windows with 4× overlap augmentation)—zero shared samples by construction.

Three results. **(R1) The domain gap is real and asymmetric.** The frozen model classifies only 66% of real WiFi windows as WiFi (best at low SNR, where region occupancy dominates; worst at high SNR, where fine-structure mismatch dominates). **(R2) The canonical compliant-attacker conclusion replicates on real signals.** With real WiFi as the victim signal, the compliant MEAP is -35.9 dB (frozen) and -37.7 dB after the leakage-free fine-tune (73% real-WiFi accuracy)—bracketing the synthetic-task value of -37.1 dB and its three-seed spread. The price of compliance is 7.3 dB (lower bound), matching the synthetic 7.1 dB. **(R3) Synthetic-only evaluation overstates robustness.** At -45 dB PSR—an attack three and a half orders of magnitude below the signal—synthetic windows flip at 0.0% while real WiFi windows flip at 67.35%: the model's correctly-classified real windows sit essentially on the decision boundary (near-zero margin), so gradient-aligned perturbations of almost any power flip them. Conditional-ASR at ultra-low power therefore measures *decision-margin pathology on out-of-distribution inputs*, not attack power; a synthetic-only evaluation hides this fragility entirely. For deployment gate purposes the actionable reading is the opposite direction: any learned coexistence sensor trained on synthetic waveforms must be red-teamed against real captures before its robustness claims are believed.

5.8. A second victim and surrogate-model transfer

A red-team suite that has only ever attacked its own author's model proves nothing about anyone else's. We therefore train a second victim, a 2-channel early-fusion ResNet (493k parameters, six times the canonical InceptionTimestyle CNN) on the identical task, data recipe, and seed (100% clean). Against it, the same compliant attack reaches 20% conditional ASR at $-25.5\,\text{dB}$ (genie $-29.2\,\text{dB}$, PoC $3.7\,\text{dB}$) versus $-37.1/-44.2/7.1\,\text{dB}$ against the canonical victim: **the victim architecture shifts the compliant-attack threshold by 11.6 dB (genie 15.1 dB) and halves the price of compliance.** Two readings follow. First, the suite has genuine discriminating power—model choice is a larger robustness lever than the emission mask itself, which is exactly the kind of statement a deployment gate needs to make. Second, no single-victim number should be quoted as "the" robustness of DL spectrum sensing; MEAP is a property of the (model, threat) pair.

Surrogate-model transfer. Crafting the attack against one model and deploying it against the other (the partial-knowledge attacker: task known, victim weights unknown; attacker channel still perfect) costs 11–23 dB (white-box minus transfer, per direction and setting): genie transfer MEAPs of $-18.4\,\text{dB}$ (dual$\to$ResNet) and $-21.0\,\text{dB}$ (ResNet$\to$dual) versus $-44.2/-29.2\,\text{dB}$ white-box; mask transfer $-10.0/-18.0\,\text{dB}$ versus $-37.1/-25.5\,\text{dB}$. The white-box worst-case disclosure that qualifies every ASR in this paper is therefore a genuine upper bound by a wide margin, and even a surrogate attacker still pays 3–8 dB of price of compliance.

5.9. Power-amplifier reality check: compliance beyond floating point

Everything above defines compliance at the attacker's transmit port, in floating point: the projection nulls out-of-mask bins exactly. A real transmitter has a power amplifier, and AM/AM nonlinearity causes *spectral regrowth*—energy reappears out of band after the projection has already happened. If the optimized adversarial waveform has high peak-to-average power ratio (PAPR), a real PA both breaks its compliance and distorts it in-band; the "regulator-verifiable adversary" claim then needs to say *where* verification happens. This section runs that check. The protocol was pre-registered before execution (hypotheses H1–H3 in `papr_pa_results.json`), following the paper's disclosure discipline.

Setup. Memoryless Rapp PA $y = x[1 + (|x|/A_s)^{2p}]^{-1/(2p)}$ with $p \in \{2, 3\}$ (softness) applied per window at input backoff IBO $= 10 \log_{10}(A_s^2/P_{\text{in}})$ swept over 0–12 dB (plus a 1 dB fine grid to 16 dB for the compliance-backoff search); output window energy is re-normalized to the attack budget (a real transmitter power-controls to its licensed limit, so compression is compensated and PSR labels stay physical). Compliance gates on the post-PA transmit spectrum use the same per-bin FFT grid as the projection: *strict* = peak out-of-mask PSD ≤ -40 dBr w.r.t. the in-band peak (802.11-style far-out level), *lenient* = -28 dBr (shoulder level); a backoff passes if 95% of windows meet the gate. Canonical urban/untargeted setup: 300 windows, attack seed 7, PGD-10, `psd_margin` 2. The in-process no-PA control reproduces the canonical MEAP at -37.0 dB (0.1 dB of the coarser canonical grid). PAPR is measured on uniform samples without oversampling (a disclosed lower bound); genie deltas behave like mask deltas and are not PA-realistic by construction, so the genie reference stays the canonical -44.2 dB.

The optimized waveform is the highest-PAPR signal on the air. The mask-constrained optimum at -36 dB PSR has mean PAPR 11.05 dB (p95 12.8), versus 6.26 (PC5), 8.77 (11p), 8.69 (WiFi), 9.02 (noise) for the benign classes—H1 confirmed. The PGD optimum exploits coherent time-domain peaks within the flat PSD cap, producing a waveform that is *harder to amplify* than every signal it impersonates.

Regrowth breaks compliance at operating backoffs. At the ∼6 dB backoff typical of OFDM transmitters, the post-PA peak out-of-mask PSD sits at -19.3 dBr (p95, p=3): 92% of windows fail even the lenient -28 dBr shoulder (all fail the strict gate), and the regrowth pedestal rises ∼9 dB above the post-PA skirt of the *cleanest* legitimate neighbor (PC5: -28.7 dBr total, of which the PA contributes only $+1.0$ dB at that backoff; 802.11p's own skirt is 6 dB less suppressed). Restoring strict-gate compliance requires IBO* = 13 dB (p=3) / 15 dB (p=2)—roughly twice the backoff a compliant OFDM radio needs for its own incremental regrowth. A floating-point-compliant attack transmitted through a PA engineered like its victims' is, in the strict sense, non-compliant and *detectable as a regrowth outlier.*

But the attack does not care—and a PA-aware attacker closes the gap. Effectiveness through the PA is essentially unchanged at every backoff: MEAP shifts of +0.6 dB (IBO 0, hard clipping), −0.1 dB (IBO 6), 0.0 dB (IBO 12–13; p=2 arms identical)—the STFT magnitude features that the attack targets are robust to envelope clipping. Pre-registered H2 (a 0.5–4 dB compliant-backoff penalty) is therefore *falsified in the attack's favor*: compliance costs the naive attacker nothing but mask cleanliness. Going further, putting the Rapp model *inside* the differentiable attack chain (project $\rightarrow$ Rapp $\rightarrow$ power-control $\rightarrow$ channel) lets the optimizer see its own PA: the PA-aware attack at IBO* is *stronger than the no-PA control* (MEAP −38.2 dB, −1.2 dB shift) while achieving 100% strict-gate compliance at the PA output (−60 dBr p95 out-of-mask)—it converges to waveforms whose intermodulation products land inside its own allocation. At IBO 0 the same optimizer instead accepts clipping (post-PA PAPR 4.95 dB) and remains non-compliant, i.e. it trades compliance for raw effectiveness when compliance is unreachable anyway. H3 (kill criterion: unreachable backoff or >6 dB penalty) is not triggered.

Reading. The compliance-constrained threat model survives the PA reality check, with two sharpened statements. First, verification must happen at the *PA output port*, not the waveform file: the naive floating-point attack fails a conformance test at operating backoffs while remaining fully effective, and PAPR/regrowth statistics are an RF-side tell against it (11.1 dB PAPR in a band whose benign signals sit at 6.3–9.0 dB). Second, that tell is not a defense: a PA-aware attacker passes it while gaining 1.2 dB. The honest conclusion is the same as the paper's other stress tests—the constraint buys detectability of *unsophisticated* attackers, not safety against the worst case the threat model is designed to disclose.

5.10. Regulatory reality check: the real ETSI EN 302 571 mask

Every result above enforces an *idealized* in-band mask: a hard OOB null plus a flat 2× in-band cap. A fair question—raised by the same reasoning as the PA reality check—is whether the price-of-compliance headline is an artifact of that self-defined shape. We therefore re-ran the canonical experiment (urban, untargeted, PGD-10, seed 7, same eval sets and budgets) with the projection replaced by the *real* European ITS unwanted-emissions template: ETSI EN 302 571 V1.2.1, Table 7 (10 MHz channel), i.e. a flat 23 dBm/MHz in-channel PSD that steps down to −3, −9, −17, −27 dBm/MHz at ±4.5, ±5, ±10, ±15 MHz offsets from the attacker's carrier, interpolated linearly in dB (values extracted from the official ETSI PDF and committed as machine-readable tables with provenance and file hash; see `data/standards/en302571_tables.json`; the archived "standard" in our earlier search corpus turned out to be a WAF block page, not the document). For the US regime we also record the C-V2X OBU unwanted-emissions limits of 47 CFR 95.3205 (−16 dBm/100 kHz within ±1 MHz of the band edge, −13 dBm/MHz to ±5 MHz, −16 dBm/MHz to ±30 MHz) in the same artifact.

The result is a wash: the ETSI-shaped compliant MEAP is −36.7 dB versus −37.1 dB for the idealized flat cap (+0.4 dB, well inside the Wilson 95% CI of $[-39.0, -35.2]$ dB), and the PoC moves from 7.1 to 7.5 dB. The direction is informative: the real template is *slightly more restrictive* for the attacker, because its in-channel edge-shaping caps the PSD near the allocation edges that the flat 2× cap leaves free, and that edge freedom was worth more to the attack than the OOB skirt the real mask permits. The conclusion is the one the PA reality check already drew, now at the mask level: the price-of-compliance physics is set by the *allocation* (where the attacker may radiate), not by the fine shape of the mask. Our idealized mask is not merely a convenience—it is (slightly) attacker-favorable, which is the conservative direction for every defense claim in this paper. Artifact: `results/etsi_mask_results.json` (ETSI curve, flat-cap reference, shape-delta, and the verbatim template knots).

5.11. From sensing error to safety harm: BSM delivery

Sensing metrics (MEAP, PoC) do not by themselves establish safety consequence. The missing chain is: false IDLE $\rightarrow$ the fooled station transmits when it should have deferred $\rightarrow$ its packet collides with a legitimate BSM at a third receiver $\rightarrow$ PRR loss. We close it with a disclosed-parameter link-budget layer anchored to the measured targeted-noise mask-ASR curve (the cloaking attack, MEAP -14.8 dB): all of FSPL anchor (47.85 dB at 1 m, 5.9 GHz), path-loss exponents (urban 3.0, swept 2.5–3.5), OBU transmit power (23 dBm), noise floor (-95 dBm, 10 MHz, NF 9 dB), decoding threshold (6 dB), capture margin (8 dB), lognormal shadowing ($\sigma = 4$ dB), and the sensing-window overlap probability (0.6, swept 0.4–0.8) are stated in the artifact JSON and swept in its sensitivity block.

Attack feasibility. At the 20% false-IDLE operating point, if the transmitter the attacker cloaks is 100 m from the sensing victim, the attacker's required EIRP stays below the 33 dBm ITS class cap out to $d_A \approx 650$ m of separation from the victim; at the 38% false-IDLE point (PSR -10 dB) it stays feasible to ≈ 450 m. The attack is physically deployable at inter-vehicle scales by a device that is, by construction, inside ITS power rules.

Harm. Figure 4 quantifies the downstream PRR. At the MEAP operating point (20% false-IDLE, synchronized worst-case timing), the legitimate BSM PRR at 100 m drops from 0.857 to 0.762 (-9.5 pp; $\sim$95 extra lost BSMs per 1000 transmitted); at 150 m, from 0.377 to 0.335; at the -10 dB operating point, 0.857$\rightarrow$0.676 at 100 m (-18.1 pp). With packet-capture modeling the numbers are within 0.2 pp. The average case matters as much as the worst case: if the attacker's energy detection is not synchronized to the packets it cloaks, the effective duty of harmful overlaps falls by an order of magnitude (10 Hz BSMs, 0.5 ms airtime) and the random-timing PRR loss at 100 m shrinks to 0.2 pp—the harm concentrates on attackers that key their waveform to the transmission they are cloaking, which energy detection makes practical.

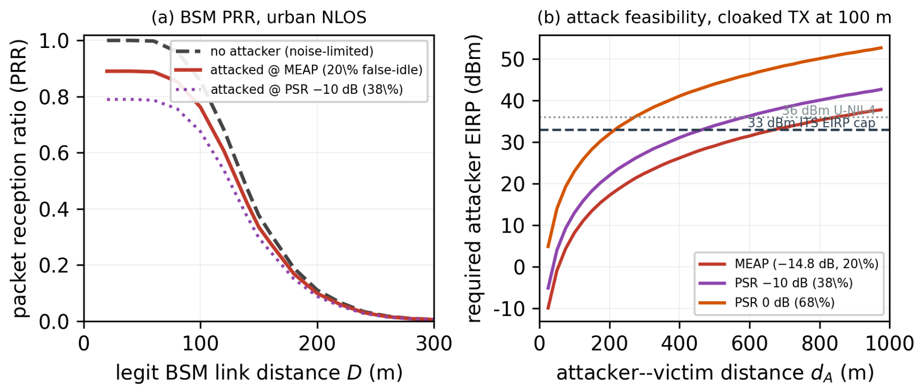

Figure 4: The harm chain, urban NLOS parameters. (a) Legitimate BSM PRR vs. link distance without and with the compliant attacker at the targeted MEAP and −10 dB operating points (synchronized worst-case timing, no-capture model). (b) Required attacker EIRP vs. attacker- victim distance with the cloaked transmitter at 100 m; the 33 dBm ITS cap makes the attack feasible at hundreds of metres.

This layer is deliberately parametric, not simulated: it is a link-budget bridge (each parameter standard-textbook, all disclosed, one Monte-Carlo seed, all curves in `results/harm_chain.json`), not an ns-3 or field campaign—those remain the honest next step, and the harm numbers should be read as order-of-magnitude quantification of the consequence, conditional on the measured sensing-level ASR.

6. Standards, Deployment Implications, and Ethics

The compliant-attacker threat model is deliberately aligned with how spectrum rules are enforced, which makes the results actionable in the O-RAN and conformance-testing contexts mapped by adversarial-ML threat analyses [11]. Five deployment scenarios, stated with their preconditions and non-claims:

- **O-RAN RIC robustness gate:** security engineers run the released compliant-attack suite against ML-based sensing xApps before deployment, requiring MEAP above a policy margin; conditional-ASR and PoC are logged as security KPIs. Single-receiver, differentiable-model scope;

no network-level guarantees.

- **FCC transition margin budgeting:** during the 5.9 GHz transition [12, 13], vendors and regulators use the compliant-ASR-vs-PSR curves as worst-case inputs when setting sensing thresholds and guard strategies for co-existence receivers. Not field measurements; non-compliant jammers are out of scope.

- **ETSI-style conformance extension:** conformance labs replay standardized mask-compliant adversarial waveforms through channel emulators into devices under test, complementing EN 302 571 unwanted-emission and blocking tests [18] with receiver-side adversarial robustness reporting. Not an established test case; not a substitute for OTA.

- **Benchmark for the sub-niche:** the released generators, channels, attack, and metrics with embedded configurations give attack/defense papers a common substrate—today every paper self-defines its threat model, blocking comparison.

- **Vendor CI regression gate:** ML teams run the mask-constrained attack nightly on frozen evaluation sets, alerting when MEAP drops or compliant ASR rises beyond tolerance, catching silent robustness regressions across model releases. Requires differentiable model export; minutes of CPU per run at prototype scale.

Responsible disclosure. The attack operates within existing regulatory power limits and cannot create spectrum violations; the released code enables defensive evaluation, and the defense section quantifies mitigation. We release all artifacts so that the threat model can be audited, extended, and defended against—the standard rationale for adversarial-ML publications.

7. Limitations

(1) Waveforms are synthetic and standard-parameterized, not OTA captures; pilots, preambles, and higher-order modulations are not modeled. (2) The

emission mask is idealized: a flat in-band cap (stricter than real in-band rules, which permit PRB-level concentration) and a hard OOB null; the $2\times \rightarrow 10\times$ cap relaxation shifts the PoC by only -0.9 dB, and re-running the canonical experiment against the real ETSI EN 302 571 V1.2.1 Table-7 template moves the compliant MEAP by only $+0.4$ dB (Section 5.10), so the allocation constraint dominates; a PRB-granular in-band model remains future work. (3) Channels are TR 37.885-inspired simplifications (first-tap-only Rician, UMa-flavored urban spread), block-fading within the window—defensible by Doppler coherence, but approximate. (4) The prototype attacker is white-box with perfect CSI and exact knowledge of the received realization (worst-case upper bound; the CSI-mismatch variant is released); the canonical numbers use zero-initialized PGD-10, while the defenses are additionally evaluated under converged adaptive attack (PGD-50 with 5 restarts, Section 5.6)—reported ASRs at PGD-10 remain lower bounds on the attack side and upper bounds on the defense side. (5) Prototype scale: undefended headline now carries three training seeds (mask MEAP spread 1.1 dB), two attack seeds, 300 windows, PGD-10 (PGD-50 and seed stability checked on key conditions: shifts $\leq$1.7 dB); the adaptive margins are replicated across three defense training seeds (spread 3.5–5.5 dB, larger than the undefended training-seed spread), while the PGD-10 defense readings and the TRADES-vs-AT comparison at PGD-10 remain seed-42 single-seed; the adaptive evaluation fixes the attack seed at 7 (5 restarts), so attack-seed variance is covered only on the undefended model. Confidence intervals are Wilson binomial on sampling noise only; model-seed variance (1.1 dB undefended, 3.5–5.5 dB defended) is a separate, additive source. (6) The phase-difference stream contributes nothing on this task; the "phase-aware" framing of our prior work is retired honestly. (7) The adversarially-trained defense's robustness is confined to its training power range (-25 to -5 dB); attacks at parity or above still succeed. The rural transfer and the PGD-10 TRADES reading are single-seed (the adaptive margins are 3-seed). (8) The real-capture study uses 96 gated bursts from three capture locations (74 eval / 22 train windows); PC5, 802.11p and the noise class remain synthetic, and the real class is 802.11 captured at

5240 MHz (U-NII-1, same OFDM family as U-NII-4), placed in the unlicensed half of the window. (9) The second-victim study shares the receiver front-end, so it varies the classifier architecture, not the receiver; transfer attacks still assume a perfect attacker channel. (10) The PA study uses a memoryless Rapp model with per-window backoff and output power control; no measured PA or transmitter chain is involved, PAPR is computed without oversampling, and the strict/lenient gates are per-FFT-bin idealizations of 802.11-style masks rather than regulatory measurement procedures (RBW, detector type). The PA-aware arm demonstrates that the threat survives this model class, not that any particular hardware realization would behave identically. (11) The harm-chain layer of Section 5.11 is a parametric link-budget analysis anchored to the measured ASR curve, not a system-level simulation: no MAC queuing, congestion control, or re-transmission modeling; single Monte-Carlo seed with $n = 4000$ per distance point; PRR figures are conditional on the stated standard-textbook parameters and should be read as order-of-magnitude consequence quantification.

8. Conclusion

Emission-mask compliance is not a defense: it shifts the adversarial attack curve on DL ITS-band spectrum sensing by only ∼5–7 dB (untargeted) and 16–18 dB (targeted cloaking), while an undefended CNN is broken by a rule-abiding transmitter operating tens of dB below the victim signal. The legacy feature-space threat model, translated to physical power, overstates realizable attacks: its own perturbation, made physical, mostly fails, while a purpose-optimized waveform at the same implied power is far stronger; the PSR-equivalence translation makes such comparisons meaningful. The power-amplifier reality check sharpens the regulatory reading: the naive floating-point-compliant attack is the highest-PAPR signal on the air and fails a post-PA conformance test at operating backoffs while losing almost no effectiveness, and a PA-aware re-optimization is both stronger and post-PA compliant—so compliance testing should sit at the PA output port, and compliance itself buys detectability of unsophisticated attackers, not safety against the worst case. The mask idealization survives contact with the real ETSI EN 302 571 template (+0.4 dB), and the harm chain closes the gap from sensing error to fleet consequence: inside ITS power rules, a synchronized attacker at the targeted 20% false-IDLE point costs 9.5 pp of BSM PRR at 100 m. Two protocol lessons generalize beyond this paper: defense evaluations that are not converged (step-starved PGD, no restarts) will systematically overstate robustness—ours did, by a factor of two—and sensing-level ASR must be paired with a link-budget layer before it can be called a safety claim. The compliant-attacker threat model, the MEAP and Price-of-Compliance metrics, and the released end-to-end framework give regulators, standards bodies, and O-RAN security engineers a physically grounded, certification-style way to measure—and, with mask-matched adversarial training, to raise—the cost of attacking deep-learning spectrum sensing: +6.7 dB against a converged adaptive attacker at zero clean-accuracy cost (AT, canonical defense seed; $+9.8 \pm 2.8$ dB across three defense seeds), +9.9 dB with a TRADES variant at 0.75 pp clean cost ($+11.5 \pm 1.8$ dB 3-seed). The stress-tests say the same thing in more ways: seed-stable, real-signal-valid, architecture-discriminating, error-barred, and now adaptive-attack certified—the numbers move, the conclusions do not.

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